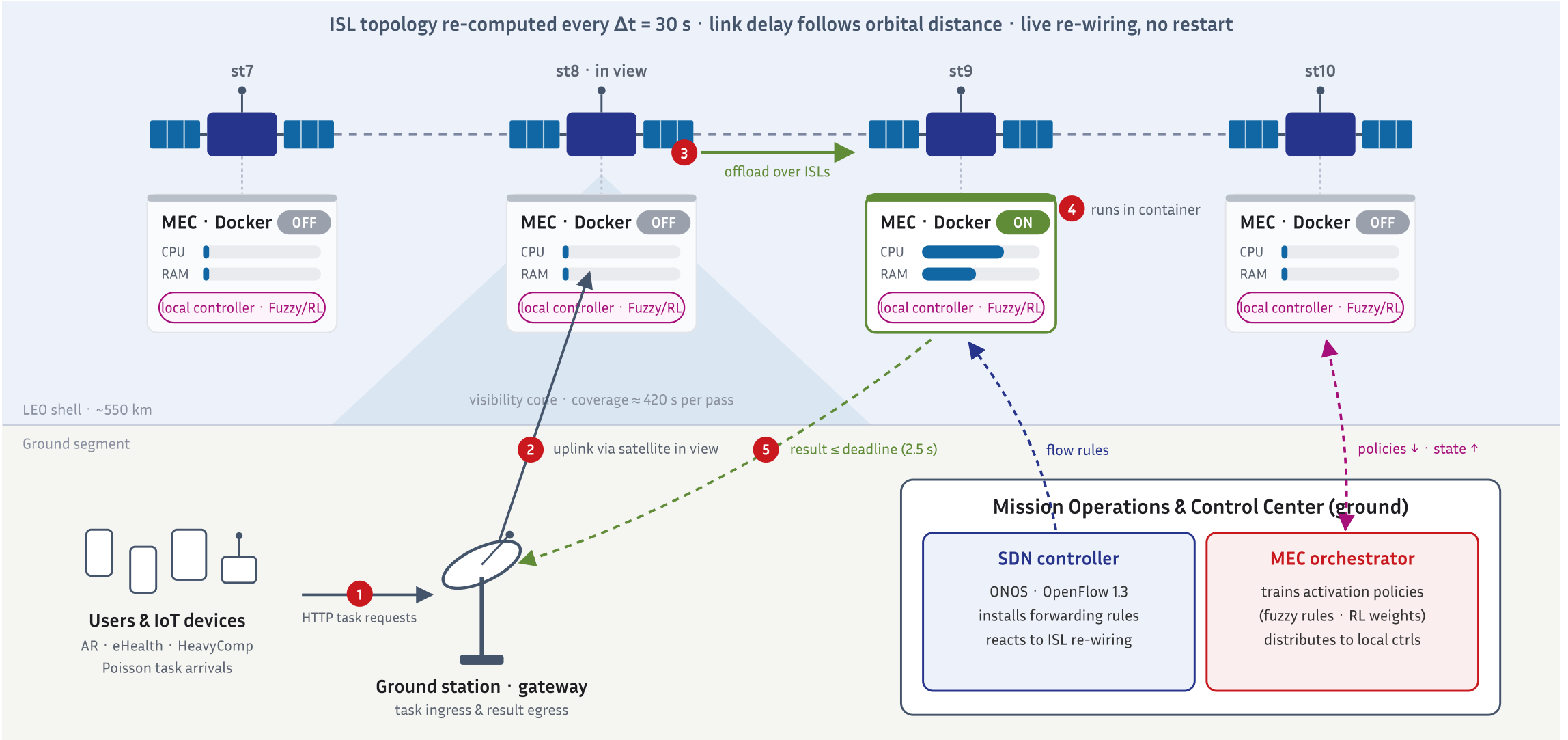

ISL topology re-computed every Δt = 30 s · link delay follows orbital distance · live re-wiring, no restart
st7
st8 · in view
st9
st10
3
offload over ISLs
MEC · Docker OFF
CPU
RAM
local controller · Fuzzy/RL
MEC · Docker OFF
CPU
RAM
local controller · Fuzzy/RL
MEC · Docker ON
4
runs in container
CPU
RAM
local controller · Fuzzy/RL
MEC · Docker OFF
CPU
RAM
local controller · Fuzzy/RL
LEO shell · ~550 km
visibility cone · coverage ≈ 420 s per pass
Ground segment
2
uplink via satellite in view
5
result ≤ deadline (2.5 s)
flow rules
policies ↓ · state ↑
Mission Operations & Control Center (ground)
SDN controller
ONOS · OpenFlow 1.3
installs forwarding rules
reacts to ISL re-wiring
MEC orchestrator
trains activation policies
(fuzzy rules · RL weights)
distributes to local ctrls
Users & IoT devices
AR · eHealth · HeavyComp
Poisson task arrivals
1
HTTP task requests
Ground station · gateway
task ingress & result egress